

Kill the Quote Spreadsheet

How I got there

The one-page note says what I decided. This appendix shows the working behind it: how I framed the problem, the dataset I built to stress it, the flows and system I designed, what the screens look like, how I measured whether any of it was true, and what I changed when the evidence said I was wrong.

1	Approach and timeline
2	Framing the problem
3	Choosing the category
4	The dataset and the buyer
5	Five vendors, five ways to break it
6	The replies, as received
7	Users and the buyer's journey
8	User flows

9	System design
10	Extraction and normalisation
11	The product, screen by screen
12	Evidence: evals, tests, the question battery
13	Auditing my own work
14	Decision log
15	What I'd build next, and honest limits

Think first, build the risky part first, then try to break it

48 hours. The order mattered more than the speed: I spent the first evening on concepts and data design before writing product code, built the part most likely to fail (extraction) before the part most likely to impress (UI), and spent the second day auditing my own prototype as a sceptical panel would.

PHASE	WHAT I DID	DECISION IT PRODUCED
Read the brief twice	Separated what is built (one flow, end to end) from what is graded (ugly edges, trust, judgement) and the invitation in the last paragraph.	Effort goes to reading, normalising and trust; drafting stays thin.
Concepts	Procurement mechanics (L1/L2, landed cost, GST, payment terms), extraction vs normalisation, human-in-the-loop, evals.	"AI reads and interprets, code calculates, the buyer decides."
Category + data	Compared four categories; designed 30 lines, a computed box weight, last year's prices, a questionnaire and five vendors with one trap each.	Corrugated packaging, with a planted twist (section 5).
Answer key first	Generated the ground truth alongside the documents: the right ₹/pc for every vendor × line and the facts a buyer must catch.	Every later claim is scored, not asserted.
Architecture	Streamlit + Python + Claude API; rejected Vercel and Netlify for a long-running Python app with 30–60 s AI calls.	One codebase, free hosting, live link for the panel.
Extraction	Readers per format, a strict schema, literal extraction, deterministic matching and normalisation.	First run 129/150 coverage exposed word-only items, which led to AI-assisted matching.
Screens and chat	Built review, compare and award; one chat panel that drafts the RFQ, then answers questions; voice input.	Review is a gate before compare; analyst answers only by calling code.
First feedback	My own walkthrough: too many tabs, no visible emails or replies, compare next to review.	One screen per step, replies shown as received, "Ask vendor" emails.
Audit + UX review	Played the Aerchain panel, a CFO and an auditor; ran a 13-question adversarial battery, mutation and failure tests, a browser walkthrough.	Found a silent failure and a wrong answer; fixed both (section 13).
Package	One-page note, this appendix, a recorded analyst walkthrough.	Show the thinking, not only the output.

The most useful hour was the one spent trying to break it.

The audit found a vendor that silently lost 21 of its 30 lines once in four runs, and an analyst answer that confidently said there were 2 five-ply lines when there are 13. Neither showed up on the happy path.

The brief describes a week; the grade is about what happens when the system isn't sure

Everyone can build the happy path. I treated the brief's three lenses as the real spec and asked of every feature: does it reduce retyping, make uncertainty visible, or make the award defensible?

What the brief actually grades

LENS	WHAT IT MEANS IN PRACTICE
Ugly edges	Angled photo, 27 of 30 lines, USD, "per box" vs "per 100". The question is what the screen shows when the reader is unsure.
Trust	Would a buyer act on ₹4 crore from this screen? Every number needs a source, a working and an honest state.
Judgement	Hundreds of calls with no right answer. Show the call, the reason and the cost.
Hard rule	Stub the plumbing; the AI loops must be real. No faked extraction, no hardcoded answers.

Where I chose to go deep vs stay thin

PART	DEPTH	WHY
Read every format	DEEP	Days are lost retyping.
Normalise + match	DEEP	Where wrong awards come from.
Review and trust	DEEP	What the panel tests.
Analyst	DEEP	The VP's question.
RFQ drafting	THIN	Any model can write an RFQ.
Email, portal, auth	STUBBED	The brief allows it.

Procurement mechanics that shaped the design

CONCEPT	WHY IT MATTERS HERE
Landed cost, not quoted price	Freight, GST treatment, discounts and conditions decide who is actually cheapest. The common basis is ₹ per piece, delivered to Bawal, before GST.
Technical before commercial	Indian buyers clear the quality questionnaire first, then open prices. The VP's question ("only among vendors who cleared") is exactly that gate.
L1 / L2 / L3	The native vocabulary: cheapest, second, third per line. The comparison shows L1 per line.
GST is recoverable	A GST-inclusive quote looks 18% dearer than it is. Compare ex-GST, show the treatment.
History matters	"Rest same as last year" is only answerable if last year's contract is in the system.

An anchor from compliance. Matching "the namkeen 150g shipper" to CB-304 is entity resolution, the same problem as screening a name against a sanctions list. The lesson carries over: a confident wrong match is worse than a flagged one, so start by over-flagging and tune with real decisions.

Corrugated packaging, because its units are genuinely messy

The category decides which ugly edges you get to show. I wanted one where the mess is real, not staged.

CATEGORY	UNIT MESS	MATCHING DIFFICULTY	REALISTIC CHANNELS	VERDICT
Corrugated packaging	Per box, per kg, per 100, per 1,000; inches vs mm	Hard: size, ply, paper grade, print	Small Indian vendors send photos and one-liners	CHOSEN
IT hardware	Low: per unit	Easy: model numbers	Formal quotes	TOO TIDY
MRO spares	Medium	Medium: part numbers, "equivalents"	Mixed	BACKUP
Freight lanes	Per trip vs per tonne, fuel surcharge	Easy: lanes	Few document types	NARROW

Why it was the right risk

- **The ₹/kg to ₹/box conversion is a perfect demo of "AI reads, code calculates"**. The model reads "₹42/kg for the 5-ply"; code multiplies by a box weight it computed from size and paper grade, and shows the working.
- **The brief's own examples point here** (₹/kg, 5-ply, "per box"), so the panel will instantly know whether the edge cases are realistic.
- **The risk:** it's the obvious pick, so others will choose it too. The answer is depth inside the category, not novelty.

Domain detail worth including. Box prices move with kraft paper rates, so serious buyers ask for price validity and a raw-material escalation clause. Both are in the questionnaire; Kraftline answered with a kraft-index clause, Pacific gave 15-day validity.

A food company buying 30 box types worth ₹4.1 crore a year

Fabricated, but built so a procurement person would nod: real specs, computed weights, last year's prices, and a total that matches the brief's “₹4 crore on the line”.

The buyer

A mid-size FMCG foods maker with a plant in Bawal, Haryana. Boxes are named after what they carry (atta, oil, biscuits, exports), because that's how buyers and plants talk about them. The category buyer runs the RFQ; a VP, a quality head and finance shape what the screen must show.

30 lines in six groups

GROUP	LINES	WHY IT'S THERE
3-ply shippers	10	High volume
5-ply shippers and tray	12	Om Sai's “₹42/kg for the 5-ply”
7-ply heavy duty	2	Specialty; some vendors skip
Die-cut	2	Hard to quote
Partitions	2	Tied to a parent box
Layer pads	2	Flat, priced by area

Box weight is computed, not invented

weight = blank area × (liner GSM + flute GSM × 1.45 take-up) × 1.04 trim

RSC blank = (2L + 2W + 35 mm glue flap) × (H + W + 10 mm)

CB-501 (330 × 250 × 290 mm, 5-ply) weighs 0.566 kg, so “₹42/kg” honestly becomes ₹23.77 per box before print and freight.

Last year is stored

FY26 contract prices (conversion ₹36/kg 3-ply, ₹40/kg 5-ply, ₹46/kg 7-ply, plus print adders) make “rest same as last year” resolvable, and savings measurable on like-for-like lines.

The questionnaire (9 questions)

GATE	QUESTIONS
Mandatory	Valid ISO 9001 · test report per lot · lead time ≤ 10 days · food-safe inks
Preferred	60-day payment · price validity ≥ 90 days
Info	Capacity · escalation clause · rejection rate

Each vendor tests one capability; no two fail the same way

VENDOR	SENDS	TRAPS	TESTS
Kraftline Manesar, large	Excel, own template	Own SKU codes, per 1,000 pcs, quotes 27 of 30 , two cheaper "value-engineered" specs, 45-day payment	Matching, missing lines, alternates
Shree Balaji Bhiwandi	Letterhead PDF + certificate	Prices include GST , mixes per kg and per piece, 3% discount in a 6pt footnote , ISO certificate expired 15 Sep	Footnotes, tax, reading attachments
Pacific Chennai, MNC	Word letter	All 30 prices in paragraphs , USD , 15-day validity, vague questionnaire	Prose extraction, FX, weak answers
Gupta Faridabad, small	Phone photo, WhatsApp	Angled rate card, inch sizes, per 100, plain-box rates, handwritten corrections , no questionnaire, no paper grade	Images, near sizes, "unknown" vs "failed"
Om Sai Gurgaon, incumbent	One-line email	"₹42/kg for the 5-ply, 38 for the 3-ply, rest same as last year , freight extra "	History, kg→box, estimating unknowns

Coverage of the brief's edge cases

EDGE CASE IN THE BRIEF	WHERE
Excel ignoring the template	Kraftline
Discount in a footnote	Shree Balaji
Commercials in a paragraph	Pacific
Angled photo	Gupta
"Same as last year, freight extra"	Om Sai
27 of 30 lines	Kraftline
USD quote	Pacific
Per box vs per 100 pieces	Gupta, Kraftline
<i>Added:</i> GST-inclusive, expired certificate, spec substitution	Balaji, Kraftline

The planted story

Random numbers make a dull demo. I designed the prices so the VP's question has a non-obvious answer:

- **Gupta looks cheapest** on raw price, but never answered the questionnaire and states no paper grade.
- **Shree Balaji looks 17% dearer** than last year, until GST is removed and the footnote applied; then it wins 19 of 30 lines against cleared vendors. But its certificate lapsed.
- **Om Sai, the incumbent**, lands mid-pack after kg→box conversion and estimated freight.

AWARD	ANNUAL COST	SAVING
Cleared vendors only	₹407.8 L	₹3.3 L
Plus Balaji, certificate renewed	₹394.9 L	₹16.1 L
Ignore quality (not advised)	₹387.1 L	₹24.0 L

The right move is "ask Balaji to renew", not "reject Balaji": about ₹12.9 L a year.

Nobody used the template

The Responses step shows each reply exactly as it arrived, before any AI touches it. Making the mess visible first is what makes the tidy comparison later believable.

Deploy

Please find attached our commercial offer against your RFQ dated 21-Sep. Rates are FOR Bawal. We have also suggested a couple of value-engineered options.

Regards,
Neha Kapoor, Key Accounts

Attachments: 1_Kraftline_Offer.xlsx

1_Kraftline_Offer.xlsx · sheet "Quotation"

0	1	2	3	4	5	6
KRAFTLINE PACKAGING PVT. LTD.						
Plot 214, Sector 8, IMT Manesar, Gurugram 122051 s						
COMMERCIAL OFFER – CORRUGATED BOXES						
Kind Attn: Procurement – Packaging, Bawal Plant				Offer No:		KPPL/Q/26
Ref: Your RFQ dated 21-Sep-2026				Date:		29-Sep-20
Our Code: Product Description: Qty: Annual Qty: Rate IN 1,000: Value IN 1: Remarks:						

1_Kraftline_Offer.xlsx · sheet "T&C + Compliance"

0	1
TERMS & CONDITIONS	
Price basis	FOR Bawal, freight included
Taxes	GST @18% extra
Payment	45 days from invoice (we are unable to accept 60 days)
Validity	90 days from offer date

Kraftline · Excel. Their own codes and layout, per 1,000 pieces, 27 of 30 lines, two “value-engineered” alternates in blue.

Deploy

Exactly as each vendor sent it. Nobody used our template.

Kraftline · Email Shree Balaji · Email Pacific · Email Gupta · WhatsApp Om Sai · Email

From: Mahesh Agarwal, Shree Balaji <quotes@shreebalaji.example> · 28 Sep 2026, 17:05

Subject: Quotation SBC/2026/447

Respected Sir,

Kindly find enclosed our best quotation along with our ISO certificate.

Thanking you,
Mahesh Agarwal

Attachments: 2_ShreeBalaji_Quotation.pdf, 2b_ShreeBalaji_ISO_Certificate.pdf

SHREE BALAJI CORRUGATORS

QUOTATION

Sr	Item Code	Description	Size (mm)	Ply	Rate (₹)	Unit
1	CB-301	Brucut 200g chipper - 1 colour flex	400x300x200	3	15.95	Per Pz
2	CB-302	Brucut 300g chipper - 1 colour flex	360x250x180	3	13.81	Per Pz
3	CB-303	Brucut 300g chipper - 1 colour flex	400x250x200	3	17.80	Per Pz
4	CB-304	Numberten 150g chipper - 1 colour flex	400x300x300	3	20.04	Per Pz
5	CB-305	Numberten 400g chipper - 2 colour flex	500x350x320	3	25.31	Per Pz
6	CB-306	Chips family pack chipper - 1 colour flex	500x400x300	3	30.81	Per Pz
7	CB-307	Samplingsheet box - 2 colour flex	250x200x150	3	7.99	Per Pz
8	CB-308	E-com master small - 2 colour flex	300x230x100	3	8.30	Per Pz
9	CB-309	E-com master medium - 2 colour flex	300x260x140	3	13.04	Per Pz
10	CB-310	Plain filler tape	250x250x250	3	13.80	Per Pz
11	CB-311	OS 1L x12 chipper - 1 colour flex	130x250x290	5	49.02	Per Kg
12	CB-312	OS 1L x4 chipper - 1 colour flex	380x340x320	5	49.02	Per Kg
13	CB-313	OS 1L per x12 chipper - 2 colour flex	340x250x290	5	50.47	Per Kg
14	CB-314	OS 1L per x24 chipper - 1 colour flex	400x300x180	5	49.04	Per Kg

CERTIFICATE OF REGISTRATION
This is to certify that the Quality Management System of
SHREE BALAJI CORRUGATORS
C-10, P-10, P-10 Industrial Estate, Bawal, Dist. JALANDHAR, Punjab, India
has been assessed and found to conform to the requirements of
ISO 9001:2015
Scope: Manufacture and supply of corrugated boxes, sheets and partitions
Certificate No: QCM/2025/0001
Date of initial registration: 14 September 2025
Date of issue: 14 September 2025
Valid until: 14 September 2028

Shree Balaji · letterhead PDF and ISO certificate. GST-inclusive prices; the 3% discount sits in a 6pt footnote; the certificate expired 15 Sep 2026.

Deploy

Exactly as each vendor sent it. Nobody used our template.

Kraftline · Email Shree Balaji · Email Pacific · Email Gupta · WhatsApp Om Sai · Email

From: Anand Krishnan, Pacific <north.accounts@pacificboxboard.example> · 29 Sep 2026, 09:18

Subject: Pacific Boxboard – proposal for Bawal

Dear Procurement Team,

Thank you for the opportunity. Our proposal is attached.

Warm regards,
Anand Krishnan

Attachments: 3_Pacific_Proposal.docx

PACIFIC BOXBOARD SOLUTIONS PVT. LTD.

(A subsidiary of Pacific Boxboard Group, Singapore)

SIPCOT Industrial Park, Sriperumbudur, Chennai 602105

Date: 29 September 2026

To: Procurement Team – Packaging, Bawal Plant

Subject: Our proposal against your RFQ dated 21 September 2026

Dear Procurement Team,

Status: Not read yet

Read all 5 replies with AI (about a minute)

Pacific · Word letter. Every price in prose, in USD; the questionnaire answered with “industry best practices”.

Deploy

1. Draft RFQ 2. Send 3. Responses 4. Review 5. Compare 6. Award

Vendor replies

Exactly as each vendor sent it. Nobody used our template.

Kraftline · Email Shree Balaji · Email Pacific · Email Gupta · WhatsApp Om Sai · Email

From: Rakesh Yadav, Om Sai <rakesh@omsaipackaging.example> · 30 Sep 2026, 21:47

Subject: Re: RFQ - Corrugated boxes FY27 (30 items)

Sir

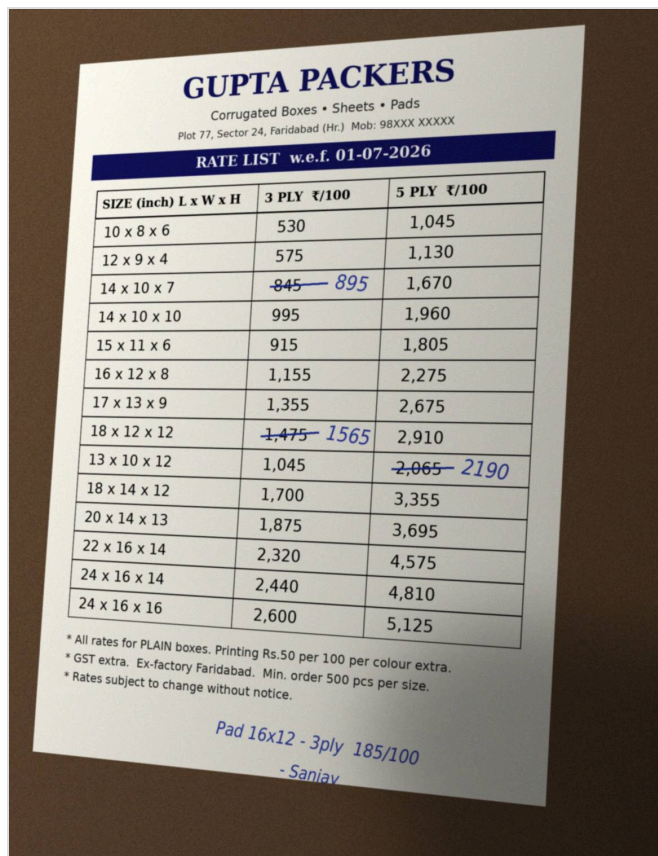
₹42/kg for the 5-ply, 38 for the 3-ply, rest same as last year, freight extra. questionnaire same as last yr.

Regards
Rakesh
Om Sai Packaging, Udyog Vihar Ph-6, Gurgaon
Sent from my iPhone

Status: Not read yet

Read all 5 replies with AI (about a minute)

Om Sai · one-line email. “₹42/kg for the 5-ply, 38 for the 3-ply, rest same as last year, freight extra.”



Gupta · the raw WhatsApp photo. Angled, inch sizes, per 100 plain boxes, three struck-through prices with handwritten corrections, and a scribbled pad rate with no ₹ sign.

What the reader recorded from this photo

FIELD	RECORDED
Prices read	29 cells, both ply columns
Units	Inches; ₹ per 100 pieces
Handwritten	4 values, struck value kept in notes
Print	Plain-box rates + "₹50 per 100 per colour" adder
Freight / GST	Ex-factory; GST extra
Pad price	Currency "unknown": not guessed
Questionnaire	Not answered (status: unknown, not failed)

The reader is told to copy, not convert. When the pad price had no currency, it wrote *unknown*. Code then assumed INR from the rest of the card and raised a review flag. Honest uncertainty beats a confident guess.

Matching his sizes to ours

GUPTA SIZE	OUR LINE	GAP	RESULT
16 x 12 x 8 in	CB-301	1.6%	MATCH
13 x 10 x 12 in	CB-501	5.1%	RISK
15 x 11 x 6 in	CB-309	8.9%	RISK
14 x 10 x 7 in, 5-ply	CB-503?	11.1%	DIFFERENT BOX

One primary user, three people who shape the screen

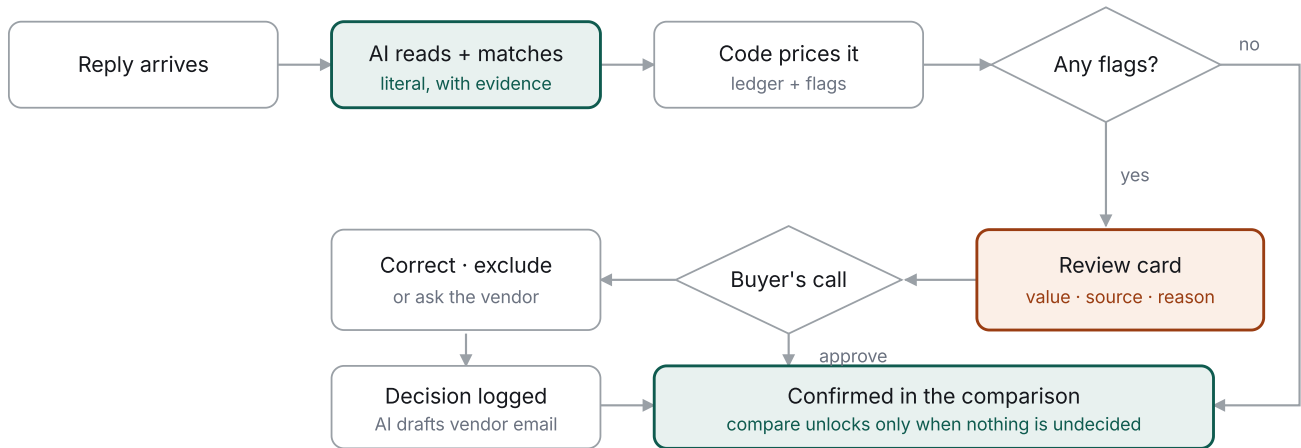
PERSON	WHAT THEY DO HERE	WHAT THEY NEED FROM THE SCREEN
Category buyer	Drafts the RFQ, reviews flags, compares, recommends the award	Stop retyping; see what's uncertain; defend every number
VP Procurement	Asks what-if questions, approves	Fast answers with the reasoning shown; savings vs last year; risks named
Quality head	Owens the questionnaire gate	Certificates and answers beside the prices; expired or vague answers caught
Finance / auditor	Signs off the commitment	Traceable working, stated assumptions, a decision record
Vendor	Replies to the RFQ	Nothing new to learn: any format, no portal

Seven stages, from “I need boxes” to a signed award

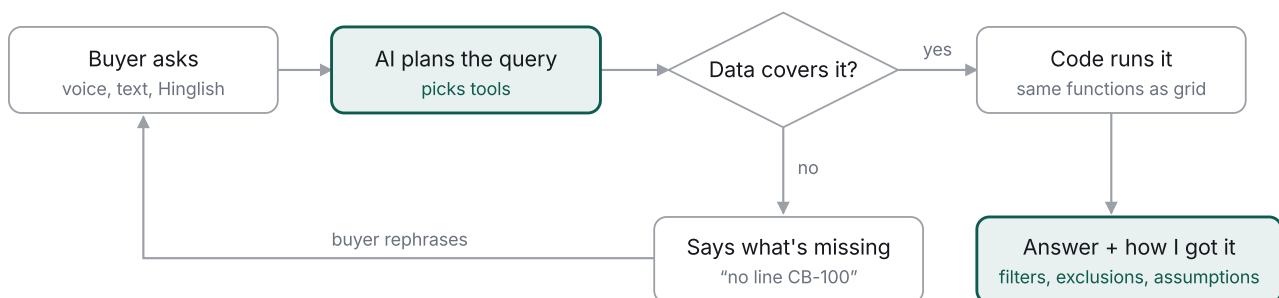
STAGE	BUYER DOES	SYSTEM DOES	BEFORE (EXCEL WORLD)
1 Draft	Speaks or types the need	Builds the RFQ through explicit edits; asks when something's missing	Copies last year's sheet
2 Send	Checks each message, sends	Email or WhatsApp per vendor, Excel attached (stubbed)	Five separate emails
3 Collect	Waits	Shows each reply as received	Chases for nine days
4 Read	Nothing	Reads all five live in parallel (~50 s), matches, converts, flags	Retypes for three days
5 Review	Clears 14 decisions, asks vendors	Groups flags; drafts vendor emails; gates compare	Doesn't know which numbers are shaky
6 Compare + ask	Scans; asks the VP's questions	L1 per line; answers by calling code	A fourth day of pivots
7 Award	Picks scope, exports	Recommendation, what's on the table, conditions, Excel	A file nobody can audit

Only flagged values reach the buyer; the analyst plans, code computes

Reviewing a vendor response



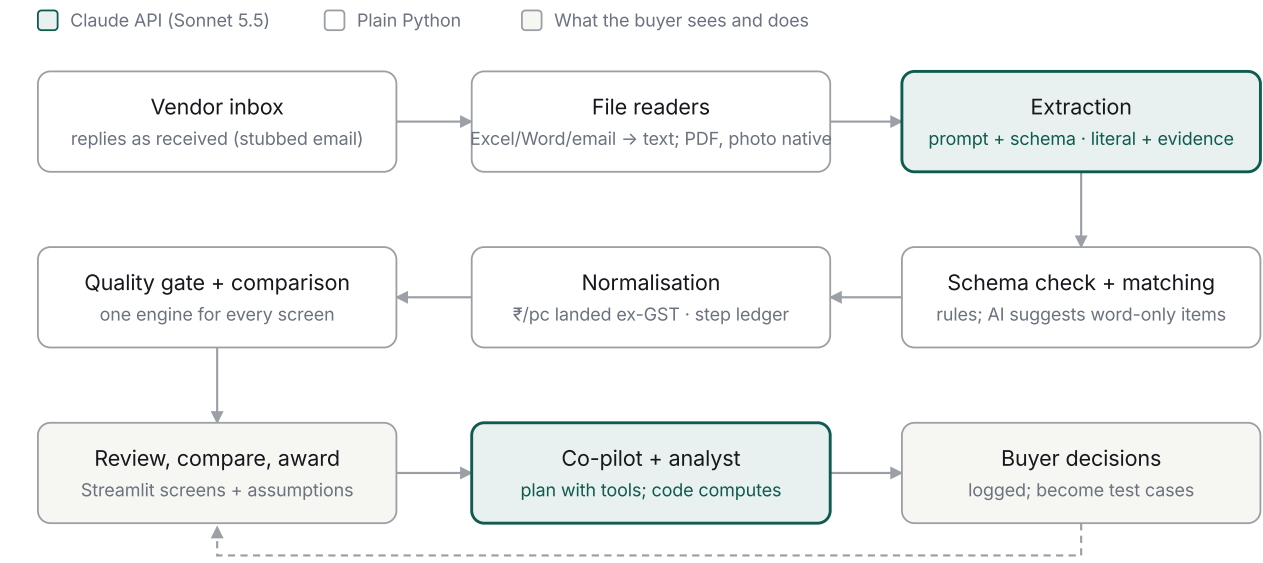
Asking a question



Why review gates compare. Comparing numbers nobody has checked is how bad awards happen. The cost is 14 clicks before the first winner appears; the gain is that every number in the comparison and every analyst answer has been seen by a human.

Why the analyst can't do arithmetic. It answers by calling the functions that draw the grid, plus a calculator. The chat and the screen cannot disagree, and a finance person can check every figure.

The model touches the data in two places; everything between is code



Stack (all free except the model)

LAYER	CHOICE
UI + hosting	Streamlit, Streamlit Community Cloud
Model	Claude Sonnet 5.5 via API (~\$10 for the build)
Validation	Pydantic schemas; malformed output rejected
Files	openpyxl, python-docx, email; PDFs and images sent natively
Voice	Browser speech recognition, typing fallback
Code	GitHub; prompts kept as plain-text files

Architecture calls

- **Not Vercel or Netlify.** They serve pages and short functions; this needs a long-running Python app and 30–60 s AI calls. In production: React front end plus a separate extraction service.
- **No fine-tuning.** Five documents aren't training data. A strict schema, domain context and an answer key do the job; corrections are logged for later.
- **A fixed pipeline, not an agent swarm.** Extraction must be explained to an auditor. Open-ended reasoning lives only in the analyst.
- **Prompts as files.** Behaviour changes without code changes, and each change can be re-scored against the key.

Copy literally, match from strict to loose, convert in code

Six stages for every file

STAGE	WHO	WHAT
1 Read	AI	PDFs and photos natively; Excel, Word, email as located text
2 Extract	AI	Value, unit, currency exactly as written, plus the vendor's words
3 Match	CODE + AI	Code → size and ply → category rules → AI for word-only items; rules veto
4 Normalise	CODE	Units, GST, FX, discounts, print, freight; every step in a ledger
5 Score	CODE	Evidence signals decide what needs review
6 Review	BUYER	Approve, correct, exclude, ask vendor

The rules that keep it honest

- **The model never converts.** If it did, a misreading and a miscalculation would look the same.
- **A near size isn't the same box.** ≤5% matches quietly; 5–10% matches as a risk; >10% is a different box. Product type must match.
- **"Not quoted" is a state,** never zero, so a 27-of-30 vendor can't win on total.
- **Only the incumbent's "same as last year" resolves.** Anyone else saying it is a risk.

One price, fully worked

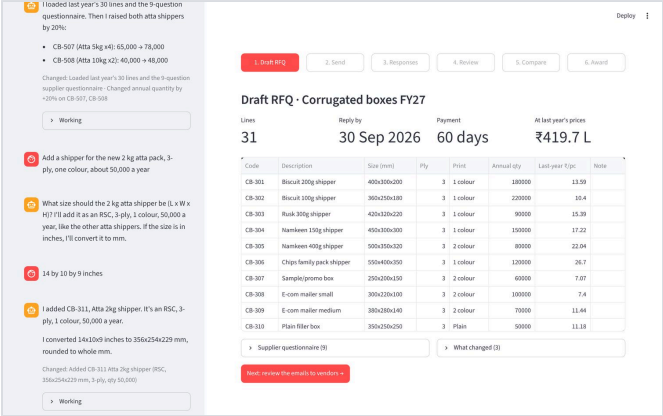
SHREE BALAJI · CB-501	VALUE
Quoted "49.02* Per Kg" (incl. GST)	₹49.02/kg
× computed box weight 0.566 kg	₹27.75
Remove 18% GST	₹23.51
Less 3% footnote discount (PO > ₹10 L)	₹22.81
₹/pc delivered, before GST	₹22.81

Confidence from evidence, not self-rating

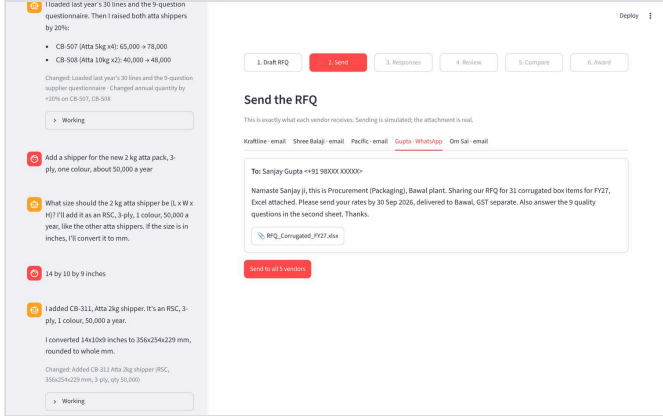
SIGNAL	BECOMES
Size gap 5–10%, value unreadable, new vendor "same as before"	RISK
Handwritten, estimated freight, assumed GST basis, conditional discount, FX, interpreted "the 5-ply"	CHECK
Per-kg converted, matched by description, paper grade not stated	INFO
More than 40% away from last year	RISK

106 → 14. Balaji's conditional discount touches 30 prices but is one decision, so it is one review card. Grouping turns 106 flagged prices into 14 calls.

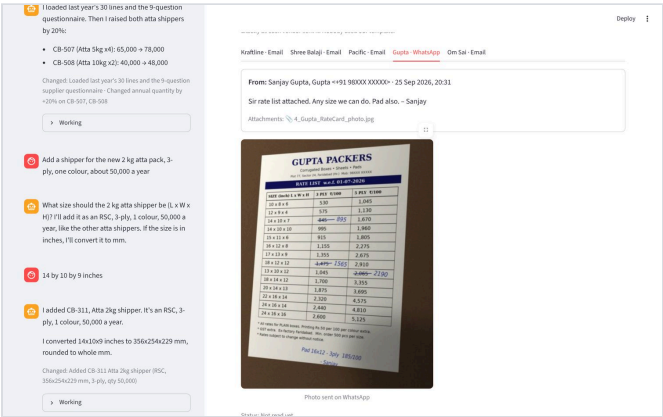
Draft, send, read



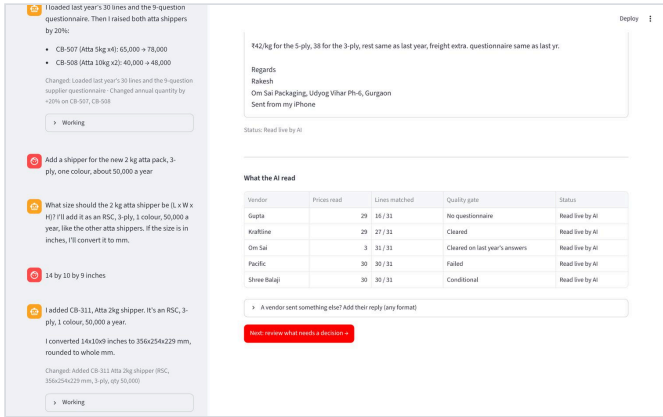
Draft by voice or text. “Bump the atta shippers by 20%” changes exactly CB-507 and CB-508. Asked for a new 2 kg shipper without a size, the co-pilot asked; given inches, it converted to mm and said so.



Send. The exact message each vendor gets: email for four, WhatsApp for Gupta, with the RFQ Excel (lines + questionnaire) attached.

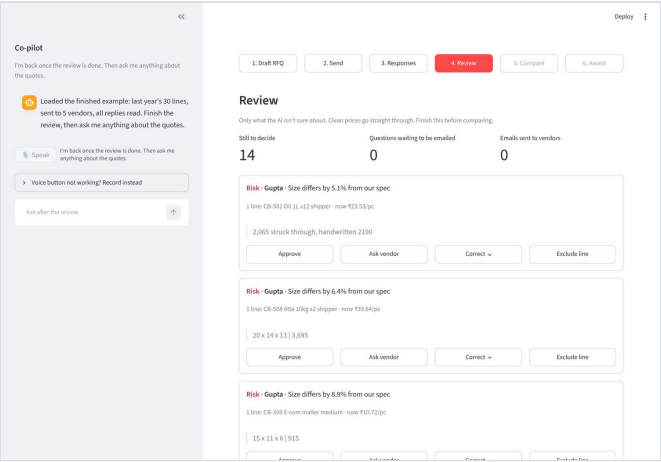


Responses. Each reply as received, including Gupta's angled photo, before the AI reads anything.

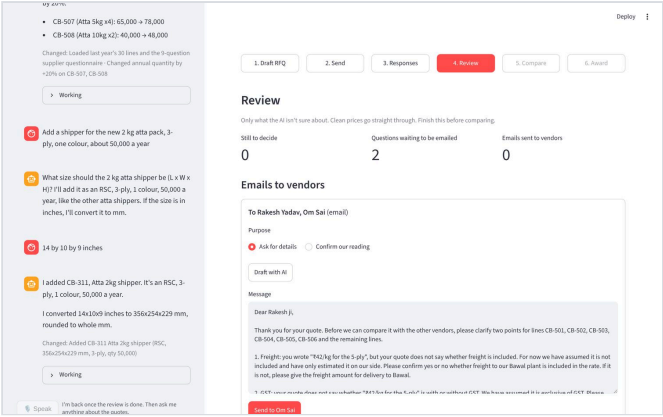


Read live. All five read in parallel in about 50 seconds; per-vendor status, lines matched and quality gate. Any other vendor's file can be added here.

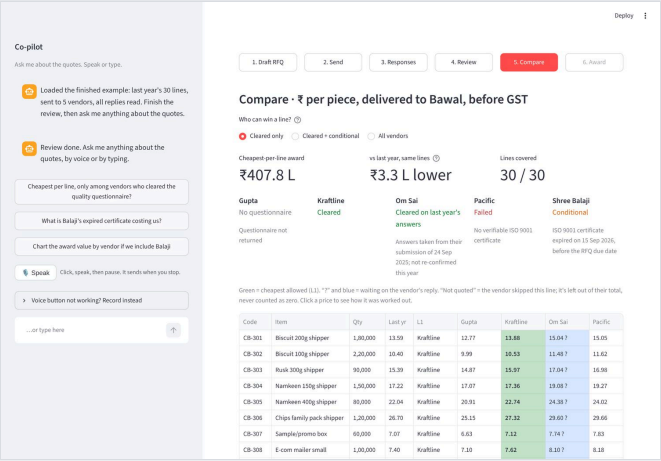
Review, compare, see the working



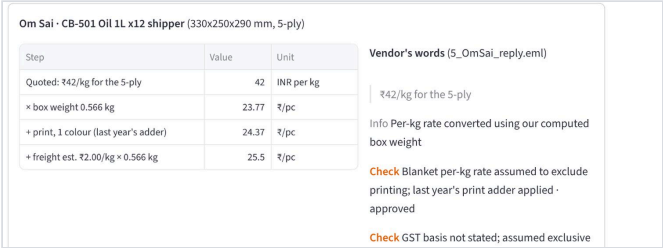
Review. Only what the AI isn't sure about, risks first, each with the vendor's own words and four actions. Compare stays locked until it's done.



Ask vendor. Open points collect into one email per vendor, drafted by the AI from the vendor's words, edited by the buyer.



Compare. ₹ per piece delivered before GST; quality gate per vendor; L1 per line; "Not quoted" in words; a toggle for who may win.



Click any price. Om Sai's "₹42/kg for the 5-ply" on CB-501: × 0.566 kg, + last year's print adder, + estimated freight = ₹25.50, with each assumption flagged.

Ask, then award

Co-pilot

Ask me about the quotes. Speak or type.

Q

Loaded the finished example: last year's 30 lines, sent to 3 vendors, all replies read. Finish the review, then ask me anything about the quotes.

Q

Review done. Ask me anything about the quotes, by voice or by typing.

Q

What if we split it, cheapest per line, but only among vendors who cleared the quality questionnaire? What's the total and the saving vs last year?

Q

Splitting each line to the cheapest quality-cleared vendor costs **₹4.08 crore a year** (₹40,780,935). That saves **₹3,26,325** (about **₹3.3 L**) against last year's **₹43,107,840** on the same 30 lines. That's under 1%.

Vendor	Lines won	Annual cost
Kraftline	26	₹38,066,555
Om Sai	4	₹2,714,360

• Om Sai's 4 lines: CB-701, CB-702, DC-101 and DC-102. Om Sai is the only cleared quote on CB-701, CB-702 and DC-101. It wins DC-102 at ₹5.67, against Kraftline's ₹5.80.

• New lines: There are none. All 30 lines existed last year, so the saving covers the whole award.

This year

Green = cheapest allowed (L1, "P" and blue = waiting on the vendor's reply. "Not quoted" = the vendor skipped this line; it's left out of their total, never counted as zero. Click a price to see how it was worked out.

Code	Item	Qty	Last yr	L1	Gupta	Kraftline	Om Sai	Pacific
CB-301	Biscuit 200g/shipper	1,80,000	13.59	Kraftline	13.77	13.86	15.04	15.05
CB-302	Biscuit 100g/shipper	2,20,000	10.40	Kraftline	9.99	10.53	11.48	11.62
CB-303	Rusk 300g/shipper	90,000	15.39	Kraftline	14.87	15.97	17.04	16.98
CB-304	Nankieen 100g/shipper	1,50,000	17.22	Kraftline	17.07	17.36	19.08	19.27
CB-305	Nankieen 400g/shipper	80,000	22.04	Kraftline	20.91	22.74	24.38	24.02
CB-306	Chips family pack/shipper	1,20,000	26.70	Kraftline	25.15	27.32	29.60	29.66
CB-307	Sample/promo box	60,000	7.07	Kraftline	6.63	7.12	7.14	7.83
CB-308	E-com mailer small	1,00,000	7.40	Kraftline	7.00	7.62	8.30	8.18
CB-309	E-com mailer medium	70,000	11.44	Kraftline	10.72	11.51	12.58	12.58
CB-310	Plain filler box	90,000	11.18	Kraftline	10.57	11.47	12.40	12.50
CB-501	Oil 1L x12/shipper	1,40,000	23.24	Kraftline	23.53	22.96	25.50	24.90
CB-502	Oil 5L x4/shipper	60,000	34.64	Kraftline	Not quoted	32.91	38.04	36.34
CB-503	Ghee 1L jar x12/shipper	90,000	22.51	Kraftline	Not quoted	21.75	24.64	23.85
CB-504	Pickle jar x24/shipper	45,000	24.82	Kraftline	Not quoted	24.47	27.22	25.52
CB-505	Sauze bottle x24/shipper	70,000	26.30	Kraftline	Not quoted	25.53	28.89	27.90
CB-506	Juice tetra x24/shipper	1,10,000	23.00	Kraftline	Not quoted	22.80	25.24	24.29

Totals row: each vendor's total on the lines they quoted, with coverage. Don't compare totals across different line counts; use the L1 column or ask the co-pilot.

Assumptions used (change one and every number updates)

Not quoted

The VP's question. Cheapest per line among cleared vendors: ₹4.08 Cr, ₹3.27 L below last year, Kraftline 26 lines, Om Sai 4, with what was excluded and why.

Time and Om Sai 1 (CB-702).

• Discount assumption: Balaji's prices include its extra 3% discount. That discount only applies on POs above ₹30 L each. If the lines are split across several POs, some may fall below that threshold and the saving would shrink.

• Quote validity: Balaji's quote is valid for only 60 days.

• Certificate: The ISO 9001 certificate expired on 15 Sep 2026, before the 30 Sep RFQ due date. If Balaji renews it, the gate may clear and this ₹12.9 L becomes available.

• Open review flags: None.

How I got this: I compared the cheapest per line split among cleared vendors only with the split among cleared plus conditional vendors (adding Shree Balaji). Gupta (no questionnaire) and Pacific (failed) are excluded in both. Prices are at ₹X.88 and freight ₹2/kg (estimated), with conditional discounts applied.

Working

Q

What's Pacific's on-time delivery rate over the last 3 years?

Q

I can't give you Pacific's on-time delivery rate. The data here has no delivery history for any vendor. It only covers this year's quotes and the questionnaire answers.

The profile does show what Pacific promised on delivery:

• Quoted lead time: 14 to 21 days from receipt of PO. *Shrinked on our production.*

This year

Green = cheapest allowed (L1, "P" and blue = waiting on the vendor's reply. "Not quoted" = the vendor skipped this line; it's left out of their total, never counted as zero. Click a price to see how it was worked out.

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Totals row: each vendor's total on the lines they quoted, with coverage. Don't compare totals across different line counts; use the L1 column or ask the co-pilot.

Assumptions used (change one and every number updates)

Not quoted

Saying "I don't know". Asked for an on-time delivery rate, it says the data doesn't exist and shows what it does have.

lines, that's under 1%.

Vendor	Lines won	Annual cost
Kraftline	26	₹38,066,555
Om Sai	4	₹2,714,360

• Om Sai's 4 lines: CB-701, CB-702, DC-101 and DC-102. Om Sai is the only cleared quote on CB-701, CB-702 and DC-101. It wins DC-102 at ₹5.67, against Kraftline's ₹5.80.

• New lines: There are none. All 30 lines existed last year, so the saving covers the whole award.

• Open review flags: None.

How I got this: I included only Kraftline (Cleared) and Om Sai (Cleared on last year's answers). I excluded Gupta (no questionnaire), Pacific (failed) and Shree Balaji (Conditional). Prices are ₹/pc delivered to Bawal before GST, at ₹X.88 and freight ₹2/kg (estimated). Conditional discounts are set to apply.

Working

Q

What is Balaji's expired certificate costing us?

Q

Keeping Shree Balaji out because of the expired certificate costs about **₹12.9 L a year** (₹1,287,903). A cleared only split comes to **₹40,780,935**. Adding Balaji gives **₹39,493,012**.

Scenario	Annual cost	Saving vs last year
Cleared only		₹12.9 L

This year

Green = cheapest allowed (L1, "P" and blue = waiting on the vendor's reply. "Not quoted" = the vendor skipped this line; it's left out of their total, never counted as zero. Click a price to see how it was worked out.

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Totals row: each vendor's total on the lines they quoted, with coverage. Don't compare totals across different line counts; use the L1 column or ask the co-pilot.

Assumptions used (change one and every number updates)

Not quoted

The insight. "What is Balaji's expired certificate costing us?" About ₹12.9 L a year, with the caveat that the 3% discount needs POs above ₹10 L.

Time and Om Sai 1 (CB-702).

• Discount assumption: Balaji's prices include its extra 3% discount. That discount only applies on POs above ₹30 L each. If the lines are split across several POs, some may fall below that threshold and the saving would shrink.

• Quote validity: Balaji's quote is valid for only 60 days.

• Certificate: The ISO 9001 certificate expired on 15 Sep 2026, before the 30 Sep RFQ due date. If Balaji renews it, the gate may clear and this ₹12.9 L becomes available.

• Open review flags: None.

How I got this: I compared the cheapest per line split among cleared vendors only with the split among cleared plus conditional vendors (adding Shree Balaji). Gupta (no questionnaire) and Pacific (failed) are excluded in both. Prices are at ₹X.88 and freight ₹2/kg (estimated), with conditional discounts applied.

Working

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What's Pacific's on-time delivery rate over the last 3 years?

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I can't give you Pacific's on-time delivery rate. The data here has no delivery history for any vendor. It only covers this year's quotes and the questionnaire answers.

The profile does show what Pacific promised on delivery:

• Quoted lead time: 14 to 21 days from receipt of PO. *Shrinked on our production.*

1. Draft RFQ

2. Send

3. Responses

4. Review

5. Compare

6. Award

Award recommendation

Who can win a line?

Cleared only

Cleared + conditional

All vendors

Annual cost

₹407.8 L

vs last year, same lines

₹3.3 L lower

Lines awarded

30 / 30

If Shree Balaji fixes the open quality issue, the award drops to ₹3.95 Cr/₹12.9 L a year cheaper. Shree Balaji: ISO 9001 certificate expired on 15 Sep 2026, before the RFQ due date. Worth asking for before you sign.

Vendor	Lines	Annual cost (₹)
Kraftline	26	3,80,66,555
Om Sai	4	27,14,360

Award value by vendor



Before signing

• Kraftline: ISO 9001 valid to 14 Mar 2028 (number stated; copy not attached)

• Kraftline: Payment 45 days instead of 60

Award. A recommendation sentence, what fixing the conditional vendor would save, the split, and conditions before signing; exports to Excel.

An answer key, built before the first line of extraction code

133/133 prices within 1% of the true ₹/pc	150/150 quoted vs not-quoted, every vendor × line	21/21 must-catch facts: footnote, cert date, handwriting, USD, alternates, terms	4/4 repeat runs of word-only matching after the fix
---	---	--	---

By vendor and format (independent live read)

VENDOR	FORMAT	PRICES	COVERAGE	FACTS
Kraftline	Excel	27/27	30/30	4/4
Shree Balaji	PDF + cert	30/30	30/30	5/5
Pacific	Word, prose	30/30	30/30	3/3
Gupta	Phone photo	16/16	30/30	6/6
Om Sai	One-line email	30/30	30/30	3/3

Is it really reading? Mutation test

I changed ₹42 to ₹39 in Om Sai's email and one cell in Kraftline's Excel (22,060 → 19,000), then re-read both. Om Sai CB-501 moved ₹25.50 → ₹23.81 (39 × 0.566 + print + freight), its 3-ply lines didn't move, Kraftline became ₹19.00, the cleared award moved ₹407.8 L → ₹403.5 L, and the analyst quoted the new number with its working.

Bad inputs degrade visibly

INPUT	BEHAVIOUR
Image of random noise	No values; "nothing legible, nothing invented"
"Sir rates same as before"	One vague item, flagged; not priced from someone else's history
Corrupt / locked PDF, empty file	Clear error: ask the vendor to resend

13 adversarial questions, checked by hand

QUESTION	RESULT
VP's split, cleared only	EXACT
Cheapest on landed cost incl. payment terms	HONEST says terms aren't costed
Looks cheapest but shouldn't be trusted	RIGHT CALL one wrong count
Every number you weren't sure about	CORRECT
Cap any vendor at 50%	MATCHES OPTIMISER 60 s
USD/INR ₹2 higher	EXACT +₹53,088
Lines with one qualified vendor	CORRECT
Draft clarification email	GOOD
Five lines for the MD + Excel	CORRECT
On-time delivery (not in data)	DECLINED
Ignore the questionnaire	COMPLIED named the risk
"Sabse sasta kaun hai 5-ply pe?"	WRONG → FIXED 2 lines → 13
Cheapest single vendor for all	CORRECT didn't rank 27/30

I reviewed it as the panel, a CFO and an auditor would

Two structured passes: a ruthless prototype audit (realness, ugly edges, trust, demo-day risk), then a UX review against an Indian buyer's journey. Each finding was labelled verified, inferred or unknown.

What the passes found

FINDING	SEVERITY	OUTCOME
One read in four: word-only matches had no line code; a vendor silently fell from 30 lines to 9	P0	Fixed: required code, retry, unmatched items shown as risk
Analyst couldn't filter by ply; wrong 5-ply answer	P0	Fixed: ply, type, size filters
Upload of a new vendor file lost in a UI rebuild	P0	Fixed
Grid printed "None" for unquoted lines	P0	Fixed: "Not quoted" + coverage totals
New vendor's "same as before" borrowed the incumbent's prices	P1	Fixed: risk, not priced
Western digit grouping, mixed dates, award with no recommendation	P1	Fixed
Savings went negative when a new line was added	P1	Fixed: like-for-like lines only
Questionnaire answers not shown beside prices; only flagged prices editable; Excel without formulas	P1	Deferred, stated in the note

Scores, before the fixes

LENS	SCORE
End-to-end flow	7/10
Ugly edges	7/10
Trust	6/10
Judgement and taste	8/10
Deliverable readiness	5/10

A wrong number that hides is worse than one that's flagged.

The matcher bug is the lesson I'd keep. Perfect scores on good runs said nothing about the bad one. The fix wasn't to make the failure rarer, but to make it loud, so the buyer sees it before it shapes an award.

Three states every value is designed for

STATE	ON SCREEN
Confident	Plain number; click for the working
Unsure	Review card with source and reason; "?" in blue while a vendor is asked
Absent	"Not quoted", never zero; coverage in totals

The calls, in the order the evidence forced them

AREA	DECISION	WHY
Principle	AI reads and interprets; code calculates; the buyer decides	Auditable numbers; clear failure modes
Scope	Deep on reading, matching, trust and the analyst; thin drafting; stubbed plumbing	Where buyers lose days; what is graded
Data	Computed box weights; stored last-year prices; answer key built with the documents	Honest ₹/kg conversion; measurable claims
Extraction	Literal values with evidence; strict schema; no conversion in the model	Separates reading errors from maths errors
Matching	Code → size/ply → category rules → AI for words; rules veto; 5% / 10% size thresholds	Suggest with AI, decide with rules
Normalising	₹/pc delivered ex-GST; ledger per price; conditional discounts flagged and toggleable; all assumptions editable	One basis; every number traceable
Trust	Evidence-based flags; 106 flags → 14 decisions; review gates compare; "Not quoted" never zero	Over-flag first, tune later
Quality	Certificates judged by date; old answers marked stale; cost of a conditional vendor shown on the award	Ask, don't reject
Loop	"Ask vendor": one AI-drafted email per vendor; lines stay marked until a reply	Surface uncertainty and act on it
Chat	One panel: co-pilot edits through tools, then analyst answers through tools; voice with typing fallback	Can't claim a change it didn't make; can't do mental maths
Flow	One screen per step in the buyer's order; replies shown as received; skip to a finished example	Simpler; the mess is visible before it's tidied
Reliability	Silent matcher failure made loud; analyst ply filters; incumbent-only "same as last year"	Found by auditing, not by luck
India fit	Lakh/crore grouping, L1 per line, dates as 15 Sep 2026, FX with stated basis	The buyer's own vocabulary
Stack	Streamlit + Claude Sonnet 5.5; no fine-tuning; prompts as files; not Vercel/Netlify	One codebase; fits long AI calls

The better problem is spec equivalence

Reading documents is close to solved. Deciding whether two vendors are quoting the same thing is not, and it is where savings leak back as damaged goods.

Next, in order

BUILD	FOR	WHY
Spec-equivalence layer	Buyer	Learn “same box” per category from every accept and reject
Questionnaire beside prices	Quality, finance	Answers and certificates one click from the number
Excel with live formulas + decision record	Finance, auditor	The comparative statement they already trust
Vendor reply merge-back	Buyer	Close the loop the “Ask vendor” email opens
Payment-term cost of money	CFO	Landed cost that includes credit days
WhatsApp intake, mobile approval	Vendors, MD	Where Indian quotes and approvals actually happen

Production path

- React front end; extraction as its own service; identity federation instead of static API keys.
- Whisper for Hinglish speech; corrections as examples; fine-tune only at volume, for cost or speed.

Honest limits

- **The data is synthetic and self-authored.** The documents and answer key were made together, so the scores prove the pipeline, not real-world accuracy.
- **The phone photo is generated.** A real, badly lit photo is the next test.
- **Shared storage on the demo host.** Fine for a demo; production needs per-tenant storage.
- **Answers can take up to a minute** for multi-step scenarios like a 50% vendor cap.
- **Payment terms, MoQ and slabs** aren't in landed cost.

Is a lighter 18 BF “value-engineered” box the same box for less, or one that crushes in the monsoon?

No OCR answers that. A system that learns the answer from every buyer's decision gets sharper with every RFQ instead of starting from zero. That is the product I'd want to build next.